Sequence coverage

Which residues of one protein were covered by an identified peptide, and what fraction of its sequence that is.

AT A GLANCE

Where	Peptide Level → Sequence Coverage
Needs	A peptide report, loaded as the Peptide level in Data → Upload, carrying ProteinNames and a stripped-sequence column. The peptide and precursor levels are read from the long-format report on disk rather than from a wide matrix, but this is the one page of this part that needs no abundance in it at all: coverage is computed from identifications, so no abundance column is chosen and no annotation is required.
Output	A monospace text map and three summary figures, not a plot. There is no plot appearance panel on this tab, because there is nothing drawn for it to style.
Denominator	The full protein sequence as it stands in a reference proteome that ships with the application — human or mouse, chosen with **Database** — and not the part of it the report knows about.
Scope	Every peptide assigned to the protein anywhere in the report. Nothing is filtered by sample, by condition or by abundance, so the map is the union over the whole experiment.
In the DataViewer	Absent. The tab is not offered in viewer mode.

1 What the map shows

Coverage answers a question the identification counts cannot: not how many peptides a protein produced, but which parts of it they came from. A protein covered at 40 % by peptides spread over its length has been sampled evenly; one covered at 40 % by peptides packed into two tryptic hot spots has three quarters of its sequence unobserved, and any modification or variant in that stretch was never in a position to be found.

The map is the protein sequence with the observed peptides written over it, so the unobserved

stretches are the runs of dashes. The percentage under the heading is the same information reduced to one number.

> **OVERLAPPING PEPTIDES DO NOT DOUBLE THE COVERAGE**
>
> On a 100-residue reference sequence, peptides covering positions 1–10 and 6–15 cover 15 distinct residues: 15 %, not 20 %. They may need separate overlay rows to remain readable. An uncovered stretch means no reported peptide matched there; it does not prove that stretch was absent from the sample.

Reading it

- A run of letters in an overlay row: one observed peptide, written at the position where it occurs in the protein sequence. A run of dashes is sequence no identified peptide covered.
- A second or third overlay row. A peptide is written into the first row whose residues at its position are still all dashes; where two peptides overlap, the second is pushed onto a new row. The rows are printed with the first one nearest the sequence line.
- The line above each block: positions, printed right-aligned at every tenth residue, so the number 50 sits over the fiftieth residue.
- The line below: the protein sequence itself, from the selected database, in blocks of ten residues separated by spaces.
- The Coverage card above the map: covered residues as a percentage of the length of that sequence, to two decimals. The Peptides card beside it is the number of distinct peptide sequences the report assigns to this protein.

> **EASY TO GET WRONG**
>
> The percentage is a fraction of length, not a score of quantification quality. The same eight peptides can give a high percentage on a 150-residue protein and a low one on a 1500-residue protein, and neither number says anything about how well either protein was quantified. Percentages can be compared as fractions, but interpret them with protein length, peptide detectability and the reference sequence. Shared peptides and repeated motifs do not uniquely locate their origin: marking every sequence match is coverage of possible positions, not proof that each copy was separately observed.

2 Controls

Controls on this page

Control	What it does	Default	Where
Database	Which reference proteome the sequence and its length come from: Human or Mouse. Both ship with the application; nothing is downloaded.	Human	page
Protein	Which protein to map. The list is built from the report: the first entry of every ProteinNames value, de-duplicated and sorted. The first of them is selected when the page opens, and the selection is reset if the report no longer offers it.	the first protein in the list	page
Chunk Size	Roughly how many residues are written per line of the map. It is never taken below 10. The map is cut every $\lfloor 1.1 \times \text{Chunk Size} \rfloor$ characters of the spaced sequence, and a ten-residue block with its separating space costs eleven characters, so 100 gives exactly ten blocks — a hundred residues — on a line.	100	page

3 How it is computed

WHICH PEPTIDES BELONG TO THE PROTEIN

ProteinNames is split on semicolons and the selected protein is looked for among the parts, so a peptide assigned to several proteins counts towards each of them. The stripped — unmodified — sequences of the matching rows are taken, de-duplicated, and used as they are.

Note that this is a wider test than the one that built the **Protein** list: the list offers only the *first* name of each ProteinNames value, while the matching accepts the name in any position.

THE SEQUENCE AND THE DENOMINATOR

The reference proteome is read as FASTA. For each entry the header is cut at its first space and split on |: the last field is taken as the entry name, the middle field as the accession. The selected protein is looked up first against the entry names and then against the accessions, as an exact string in both cases.

The sequence found is the denominator. Coverage is a fraction of the *whole* protein as that file records it, not of the part the report mentions and not of a tryptic subset of it.

COVERAGE

A mask of zeros, one per residue of the protein sequence, is set to one everywhere a peptide occurs. Each peptide is searched for as a literal substring, and *every* occurrence is marked, not only the first, so a repeated motif is covered wherever it appears. Overlapping peptides set the same positions to one and are therefore counted once. Then

$$\text{coverage} \;=\; 100 \times \frac{\#\{\, i \,:\, \text{mask}_i = 1 \,\}}{L},$$

with L the length of the sequence, rounded to two decimals.

THE MAP

The map is built separately from the percentage, by writing the peptides into overlay strings of dashes rather than into a mask. Each peptide is placed, at each position where it occurs, into the first overlay row whose residues at that position are still all dashes; if none is free a new row is opened. Rows therefore accumulate only where peptides overlap, and their number is the deepest overlap in the protein.
The sequence, the overlays and the position line are then split into blocks of ten residues joined by single spaces, and cut into lines of $\lfloor 1.1 \times \text{Chunk Size} \rfloor$ characters. Each block of the output prints the position line, then the overlay rows in reverse order so that the first-placed peptides sit closest to the sequence, then the sequence line, then a blank line.

Coverage is deliberately an overall figure: every peptide identified anywhere in the report counts, with no sample, condition or abundance filter, so the number describes the experiment rather than a run.

Human and mouse are the reference proteomes the tool provides, and a protein absent from them cannot be drawn here.

Edge cases

- A peptide that does not occur verbatim in the database sequence contributes nothing and is not reported. This is what happens when the report's assignment names an isoform or a variant that the bundled proteome does not contain.
- If the protein carries no peptides in the report, nothing is drawn and the page says so by name.
- If the selected protein is in the report but in neither the entry names nor the accessions

of the selected database, the page reports that the variant does not exist in the loaded database. Switching **Database** between human and mouse is the usual cause.

- If the database file itself cannot be found the page says which species it was looking for; the locations it searched go to the server log rather than to the screen.
- If the report has no ProteinNames column the protein list is empty, the page says that sequence coverage is unavailable for this report, and the other tabs of this part are unaffected.
- Coverage does not distinguish a protein seen once from one seen in every run: a peptide identified in a single sample covers its residues exactly as one identified in all of them does.

4 The summary above the map

The three cards

Column	What it holds
Protein	The protein as selected, spelled as the report spells it.
Coverage	Covered residues as a percentage of the length of the database sequence, to two decimals.
Peptides	The number of distinct stripped sequences the report assigns to this protein — the peptides that went into the map, whether or not each was found in the sequence.

5 Exporting

The notebook download beside **Download TXT** reconstructs the calculation and its text output with the required reference resources. It does not invent a separate image plot. Keep the reference sequence identity with the result: changing an isoform or database sequence can change both positions and the coverage denominator.

There is no figure to download and no image format to choose. **Download TXT** saves the map exactly as it is shown, as a plain text file named for the protein and the database, in the form PROTEIN_species_coverage.txt. It is monospaced by construction, so it only lines up in a monospaced font; pasted into a proportional one the overlay rows no longer sit over the residues they refer to. Neither the map nor the percentage is carried into the summary report,

which covers the other three tabs of this part only.

6 Further reading

The UniProt help page *Sequence processing* explains canonical precursor and mature sequences. Coverage depends on the actual reference sequence selected; an accession, isoform and sequence version should accompany a reproducible map.

Behind this page

Function	What it produces
calculate_coverage	The percentage: covered residues over the length of the database sequence
vis_coverage	The text map, with its overlay rows and position lines
fasta_to_dataframe	Reads a FASTA database into headers, sequences, entry names and accessions
peptide_sequence_coverage	The page's own call: the percentage, the peptide count and the map together